

Employment

Graduate Research Assistant NC State University Apr 2025 – Present

- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern Isomer AI May 2025 – May 2026

- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments.
- Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for customer-internal browser automation.

Software Developer Intern Mphasis (HP Company) Jun 2024 – Aug 2024

- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern Capgemini Jun 2023 – Aug 2023

- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC North Carolina State University Aug 2024 – May 2026

- Master of Science in Computer Science; GPA: 3.5/4.0. Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India Mumbai University Aug 2020 – May 2024

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

KubeServe Kubernetes LLM Inference Platform 2025

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability stack; 88 pytest tests.

PulseRank Real-Time Leaderboard & Activity Feed 2026

- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js; MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE.
- Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.

AlgoPulse Full-Stack Agentic AI Platform 2025

- Built agentic AI orchestration platform on Temporal.io and Kafka: agent loop with multi-step reasoning, tool invocation, and retry logic; sustained 2,000 async inference requests per second.
- Deployed on AWS with auto-scaling groups; k6 load tested to 10,000 concurrent users; sub-100ms leaderboard latency.
- Designed event-driven SSE delivery via Supabase Edge Functions; replaced polling and cut active server connections by 60%.

Languages and Technologies

- Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- Frontend & Runtime:** React, Next.js, NestJS, Node.js, Express, FastAPI, Auth0, REST APIs
- Data & Storage:** PostgreSQL, MySQL, MongoDB, Redis, DuckDB, Kafka, Testcontainers
- Infra & DevOps:** Kubernetes, Docker, AWS, GCP, OpenTelemetry, Prometheus, Grafana, GitHub Actions, ArgoCD, Helm

Additional Experience and Awards

- IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniadiitya.com/blog.